

ISHA PATRO

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EDUCATION

NEW YORK UNIVERSITY
Master of Science in **Financial Engineering**
GPA: 3.6/4.0

Brooklyn, NY
Expected 05/27

VELLORE INSTITUTE OF TECHNOLOGY, AP
Bachelor in **Computer Science**
GPA: 3.5/4.0

Andhra Pradesh, India
07/2

PROGRAMMING & TECHNICAL SKILLS

Programming & Development: Python, JavaScript, VBA, React, REST APIs, Streamlit, Dash, AG Grid, Tableau
Data Science: Statistical Modeling, Machine Learning, Deep Learning, Attention Mechanisms, LLMs, Generative AI, RAG, AI Agents, Agentic AI, MCP, NLP, Fine-Tuning, AI Inference, LLM Evaluation, Prompt Engineering
Databases & Cloud: SQL, q/kdb+, AWS, MongoDB, Microsoft Excel, Bloomberg Terminal
Quantitative Finance: Fixed Income, Portfolio Management, Risk Management, Time Series Forecasting, Prediction Markets

EXPERIENCE

FINANCIAL ARTIFICIAL INTELLIGENCE WORKING GROUP 07/26 - 09/26
Quantitative Finance Research Intern

- Developed a corpus for Fixed Income Modelling, Credit Card Securitization, and Derivatives workflows to benchmark AI agent outputs against Excel-based models, improving calculation accuracy, reasoning traceability, and hallucination resistance.

J.P MORGAN CHASE & CO. | Commercial & Investment Bank 01/21 - 08/25
Software Developer Engineer - II, MBS Division - Fixed Income Pricing Team (Bangalore, India)

- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting \$5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

RESEARCH & OPEN SOURCE CONTRIBUTION

Modeling Volatility in Polymarket Prices with Binary Jump-Tweedie Dynamics 07/26 - 08/26
Under review for the Prediction Markets Conference, Yale School of Management & Swensen Asset Management Institute

- Modeled market volatility using arcsine probability geometry, Tweedie likelihood estimation, market-level clustering, & category-specific parameter selection, benchmarking performance against GARCH and Hidden Markov baselines.

FinOKF: Evidence-Aware Finance Open Knowledge Formatting with LRU Caching for Auditable Equity Research 06/25 - 09/25

Under review at Journal of Association for the Advancement of Artificial Intelligence

- Architected an evidence-validated LLM system for equity research over SEC filings that cut token usage 97% and latency 16x across GPT-5.5, Claude, and Llama while preserving numerical accuracy on audited financial calculations.

Implied Prepayment Surfaces via Finance-Informed Neural Networks in Agency MBS Markets 11/25 - 03/26
Under review at Journal of Computational Finance.

- Built a finance-informed neural network that inverts agency TBA prices into a market-implied prepayment (CPR) surface under a Hull-White no-arbitrage constraint, showing the implied-realized divergence is a predictor of forward TBA returns

qutePandas (Open Source Contribution) 06/25 - Present

- Published a high-performance Pandas-compatible execution layer backed by kdb+/q, achieving order-of-magnitude speedups (10x-100x+) on large-scale analytical workloads while preserving Python-first research ergonomics.

Unlocking Financial Wisdom: Building Knowledge Graphs from Investopedia 06-24 - 08/24
Published in International Journal of Research and Analytical Reviews (IJRAR), Sep 2024

- Built a financial knowledge graph from 6,000+ Investopedia terms using data extraction and relationship mapping, enabling semantic visualization in Obsidian; presented at PitchWell Season 3 Finals at J.P. Morgan

HONORS / AWARDS

- Awarded the prestigious **Numerix Women in Finance Scholarship**, a \$20,000 honor, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (\$1,200) - across all engineering disciplines, 2017-201

ACADEMIC PROJECTS

Spline-Based Conditional Market Beta & Statistical Arbitrage Framework 09/26

- Modelled a NumPy-based cubic spline regression and k-fold cross-validated truncated basis engine to capture state-dependent non-linear asset co-movements, extracting idiosyncratic residual spreads for market-neutral statistical arbitrage.

Money Talks 09/26

- Engineered an AI-powered SaaS analytics platform that surfaces revenue concentration risk, tracks high-value customer churn, and evaluates go-to-market performance alongside underlying monthly recurring revenue (MRR) health.

<u>2WIN</u>	06/26
Designed and backtested quantitative prediction-market strategies using tick-level order-book data, dynamic risk-based sizing, automated execution logic, and portfolio PnL attribution across real-time Polymarket markets.	
<u>VORTEX</u>	05/26
Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.	
<u>Deep Temporal Models With Attention and Skip Connections</u>	03/26
Benchmarked 4 deep temporal architectures (RNN, BiRNN, LSTM, and BiLSTM) incorporating attention mechanisms and skip connections using FRED macroeconomic data to predict directional movements in wheat commodities.	
<u>Geometry of Intelligence</u>	01/25 - Present
Created an independent machine learning research series explaining algorithms from first principles through mathematical intuition, probability, and clean code implementations.	
<u>Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation</u>	01/26 - 03/26
Analyzed cross-currency basis and liquidity fragmentation in crypto markets using high-frequency data across exchanges, evaluating stablecoin-driven pricing inefficiencies and regulatory impact (GENIUS Act)	
<u>Monte Carlo Pricing of European Barrier Options</u>	10/25
Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.	
<u>Fixed Income Analytics & Securitization Suite</u>	09/25
Built a financial modeling suite in Excel featuring Yield Curve Calibration using Nelson-Siegel model, Mortgage-Backed Securities (MBS) delta-neutral hedging, and multi-tranche credit card ABS cash flow waterfalls.	
<u>Advanced NASDAQ Market Prediction System with Deep Learning Optimization</u>	08/25
Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.	
<u>Minted Wealth</u>	03/25
Developed a comprehensive financial intelligence dashboard integrating live market data, portfolio performance analytics, and net worth insights for smarter investment decisions.	